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Agri Twin – Digital Twin Platform for Precision Farming and Crop Health Prediction

TEST CASE DESIGN

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CSA1012 – Software Engineering

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1. Introduction

1.1 Background

Agriculture worldwide faces mounting pressure from climate change, unpredictable weather, soil degradation, pest infestations and water scarcity, all of which threaten crop yields and farmer livelihoods. Traditional farming practices, which rely heavily on manual observation and experience-based decisions, are increasingly insufficient to manage these compounding risks at scale.

Precision agriculture addresses this gap by combining IoT sensors, drones, satellite imagery and artificial intelligence to continuously monitor field conditions and optimize resource use. Digital transformation of agriculture is therefore no longer optional; it is a necessary step toward sustainable, climate-smart and data-driven farm management.

1.2 AgriTwin Overview

AgriTwin is proposed as an intelligent Digital Twin platform that builds a continuously updated virtual representation of agricultural fields by integrating real-time and historical data from IoT sensors, drones, weather stations and satellite imagery. AI-powered predictive models within AgriTwin estimate crop health, disease likelihood and pest-outbreak risk, while a recommendation engine translates these predictions into actionable guidance for irrigation, fertilization and pest management. Visualization dashboards and predictive reports support farmers in making timely, data-driven decisions.

1.3 Purpose of Testing

Because AgriTwin integrates multiple heterogeneous data sources with AI-based prediction and recommendation logic, it is exposed to a wide range of potential failure points: missing or malformed sensor data, delayed synchronization, prediction-service unavailability and unauthorized access, among others. Systematic software testing is therefore essential to verify that the platform behaves correctly under both normal and abnormal conditions, protects farm data, and produces recommendations that are traceable to valid inputs.

1.4 Testing Objectives

- Verify functional correctness of all core AgriTwin modules.
- Verify correct handling of sensor data, including invalid and missing values.
- Verify the crop-health, disease and pest-prediction workflow.
- Verify accuracy and consistency of recommendation generation.
- Verify correct dashboard visualization behaviour.
- Verify predictive report generation and export functionality.
- Verify security and reliability of the platform under fault conditions.
- Verify system behaviour under invalid and exceptional conditions.
- Ensure traceability between requirements, test scenarios and test cases.

1.5 Scope of Testing

- User authentication and session management
- Farm and field management
- IoT sensor data ingestion and validation
- Environmental monitoring (soil moisture, nutrients, irrigation status)
- Digital Twin synchronization
- Crop health, disease and pest-outbreak prediction
- Irrigation, fertilization and pest-management recommendations
- Dashboard visualization
- Predictive report generation
- Data validation, security, reliability and usability

2. System Overview and Architecture

AgriTwin is organized into layered modules that progressively transform raw agricultural data into actionable, farmer-facing recommendations. Field-level data is first acquired from IoT soil sensors, weather stations, drones and satellite imagery, then validated and used to update a Digital Twin — a continuously synchronized virtual representation of the physical field. The Digital Twin, combined with historical records, feeds an AI/ML prediction engine that estimates crop health, disease and pest risk. These predictions drive a recommendation engine that produces irrigation, fertilization and pest-management guidance, which is finally surfaced through dashboards and predictive reports.

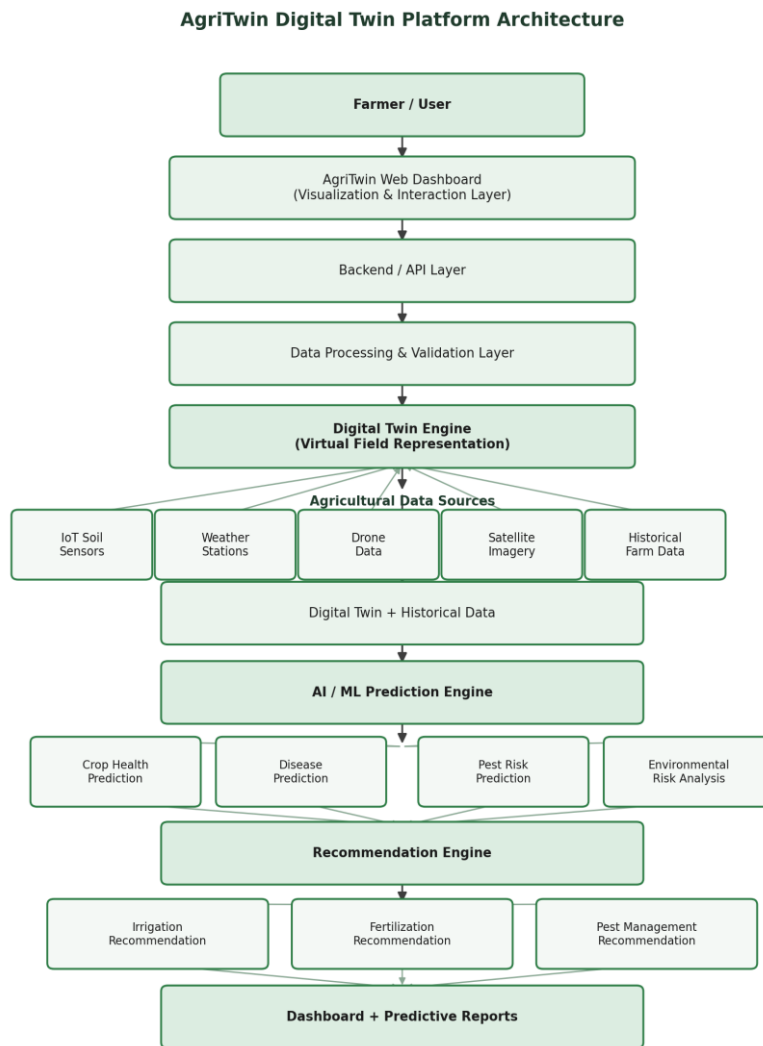


Figure 1: AgriTwin Digital Twin Platform Architecture

The end-to-end workflow below illustrates how a single unit of field data moves through the platform, from acquisition to farmer-facing output, including the exception path taken when incoming data fails validation.

AgriTwin End-to-End Workflow

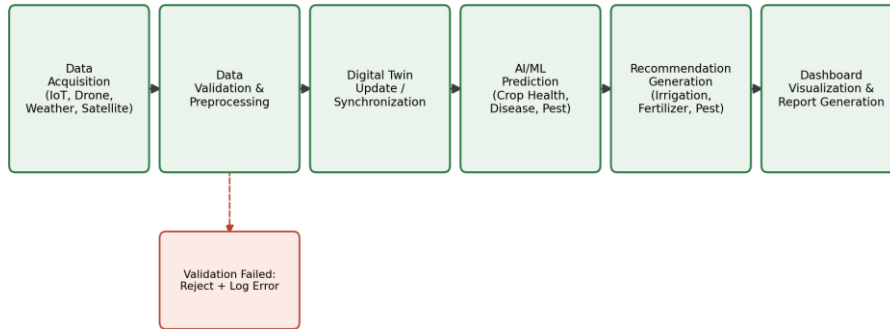


Figure 2: AgriTwin End-to-End Workflow

3. Functional Requirements

The functional requirements below were derived directly from the AgriTwin problem statement and objectives, and form the basis for test scenario and test case design in later sections.

Req. ID	Functional Requirement	Module	Priority
FR-01	User Authentication	Authentication	High
FR-02	Farm Registration	Farm Management	High
FR-03	Field Management	Farm Management	High
FR-04	IoT Sensor Data Collection	Data Acquisition	High
FR-05	Weather Data Collection	Data Acquisition	Medium
FR-06	Drone/Satellite Data Integration	Data Acquisition	Medium
FR-07	Real-Time Data Validation	Data Processing	High
FR-08	Digital Twin Synchronization	Digital Twin Engine	High
FR-09	Soil Moisture Monitoring	Environmental Monitoring	High
FR-10	Nutrient Monitoring	Environmental Monitoring	Medium
FR-11	Irrigation Status Monitoring	Environmental Monitoring	High
FR-12	Environmental Condition Monitoring	Environmental Monitoring	Medium
FR-13	Crop Health Prediction	AI Prediction Engine	High
FR-14	Disease Prediction	AI Prediction Engine	High
FR-15	Pest Outbreak Prediction	AI Prediction Engine	High
FR-16	Precision Irrigation Recommendation	Recommendation Engine	High
FR-17	Fertilization Recommendation	Recommendation Engine	Medium
FR-18	Pest Management Recommendation	Recommendation Engine	Medium
FR-19	Farm Performance Dashboard	Visualization	High
FR-20	Crop Growth Visualization	Visualization	Medium
FR-21	Predictive Report Generation	Reporting	High
FR-22	Alerts / Notifications	Reporting	Medium

4. Non-Functional Requirements

Non-functional requirements address the qualities AgriTwin must exhibit beyond individual features, with particular emphasis on real-time data reliability, sensor-data integrity, prediction responsiveness, protection against unauthorized access, correct handling of incorrect/incomplete data, system availability, scalability across multiple farms, and privacy of farm information.

ID	Requirement	Description	Testing Objective
NFR-01	Performance	The platform shall process sensor data and return predictions within an acceptable response time under normal load.	Verify prediction and dashboard response times.
NFR-02	Security	The platform shall restrict access to authenticated and authorized users and protect farm data from unauthorized access.	Verify authentication, authorization and access-control enforcement.
NFR-03	Reliability	The platform shall continue functioning correctly during sensor, network or service disruptions and recover gracefully.	Verify behaviour under sensor/API/network/database failure and recovery.
NFR-04	Availability	The platform shall remain accessible to farmers for monitoring and decision-making with minimal downtime.	Verify system accessibility across sessions and after service restarts.
NFR-05	Usability	Dashboards, alerts and recommendations shall be presented in a clear, farmer-understandable manner.	Verify clarity of visualizations, alerts and recommendation text.
NFR-06	Scalability	The platform shall support increasing numbers of farms, fields and sensor streams without functional degradation.	Verify functional behaviour with multiple farms/fields and concurrent data streams.
NFR-07	Data Integrity	Sensor and prediction data shall remain accurate, consistent and free from unintended duplication or loss.	Verify duplicate handling, data validation and consistency between twin and source data.
NFR-08	Maintainability	The platform's modules shall be structured to allow updates to prediction models and recommendation logic with minimal impact.	Verify modular behaviour is testable in isolation (e.g., prediction engine vs. recommendation engine).
NFR-09	Compatibility	The platform shall function correctly across supported browsers/devices used to access the dashboard.	Verify dashboard rendering and functional behaviour across supported access modes.
NFR-10	Privacy	Farm-specific and location data shall not be exposed to unauthorized users or other farmers.	Verify data isolation between different farmer accounts.

5. Test Scenario Identification

Test scenarios were identified by systematically walking through each functional and non-functional requirement and considering positive, negative, boundary, exceptional, security and reliability conditions relevant to AgriTwin's data-driven, sensor-integrated nature.

Scenario ID	Req. ID	Module	Test Scenario	Type	Priority
TS-01	FR-01	Authentication	Valid login with correct credentials	Positive	High
TS-02	FR-01	Authentication	Invalid login with incorrect password	Negative	High
TS-03	FR-01	Authentication	Login attempt with empty credentials	Negative	Medium
TS-04	FR-01	Authentication	Unauthorized access to protected page without login	Security	High
TS-05	FR-01	Authentication	Session expiry after inactivity	Exceptional	Medium
TS-06	FR-02	Farm Management	Add a valid new farm record	Positive	High
TS-07	FR-02	Farm Management	Add farm with invalid/incomplete data	Negative	High
TS-08	FR-02	Farm Management	Add duplicate farm record	Boundary	Medium
TS-09	FR-03	Farm Management	Update existing farm/field details	Positive	Medium
TS-10	FR-03	Farm Management	Delete an existing farm/field	Positive	Medium
TS-11	FR-04	Sensor Data	Ingest valid sensor reading	Positive	High
TS-12	FR-04	Sensor Data	Handle missing sensor reading	Negative	High
TS-13	FR-04	Sensor Data	Handle out-of-range sensor value	Boundary	High
TS-14	FR-04	Sensor Data	Reject invalid data format	Negative	Medium
TS-15	FR-04	Sensor Data	Detect duplicate sensor reading	Boundary	Medium
TS-16	FR-04	Sensor Data	Handle delayed sensor reading	Exceptional	Medium
TS-17	FR-04	Sensor Data	Handle sensor disconnected event	Exceptional	High
TS-18	FR-08	Digital Twin	Correct data synchronization to twin	Positive	High
TS-19	FR-08	Digital Twin	Delayed synchronization handling	Exceptional	Medium
TS-20	FR-08	Digital Twin	Synchronization with missing data	Negative	High
TS-21	FR-08	Digital Twin	Synchronization with conflicting data	Boundary	Medium
TS-22	FR-08	Digital Twin	Historical data integration into twin	Positive	Medium
TS-23	FR-13	AI Prediction	Prediction generated with valid input data	Positive	High
TS-24	FR-13	AI Prediction	Prediction attempt with missing input	Negative	High
TS-25	FR-13	AI Prediction	Prediction with insufficient historical data	Boundary	Medium
TS-26	FR-14	AI Prediction	Disease prediction service response handling	Positive	High
TS-27	FR-13	AI Prediction	Prediction service unexpected failure	Exceptional	High

Scenario ID	Req. ID	Module	Test Scenario	Type	Priority
TS-28	FR-16	Recommendation	Normal irrigation recommendation generated	Positive	High
TS-29	FR-16	Recommendation	Recommendation under high soil-moisture condition	Boundary	High
TS-30	FR-16	Recommendation	Recommendation under low soil-moisture condition	Boundary	High
TS-31	FR-17	Recommendation	Nutrient-deficiency fertilization recommendation	Positive	Medium
TS-32	FR-18	Recommendation	Pest-risk management recommendation	Positive	Medium
TS-33	FR-19	Dashboard	Correct dashboard visualization of farm data	Positive	High
TS-34	FR-19	Dashboard	Dashboard display with empty/no data	Negative	Medium
TS-35	FR-19	Dashboard	Dashboard shows incorrect/mismatched data	Negative	Medium
TS-36	FR-19	Dashboard	Dashboard real-time data update	Positive	High
TS-37	FR-20	Dashboard	Multiple field visualization on one dashboard	Positive	Medium
TS-38	FR-21	Reports	Generate predictive report for valid date range	Positive	High
TS-39	FR-21	Reports	Generate report with empty underlying data	Negative	Medium
TS-40	FR-21	Reports	Generate report with invalid date range	Negative	Medium
TS-41	FR-21	Reports	Generate report for a large date range	Boundary	Medium
TS-42	FR-21	Reports	Download/export generated report	Positive	Medium
TS-43	NFR-02	Security	Unauthorized access attempt to restricted module	Security	High
TS-44	NFR-02	Security	Access with invalid/expired session token	Security	High
TS-45	NFR-02	Security	Role-based access restriction enforcement	Security	High
TS-46	NFR-02	Security	Malicious input submitted to input fields	Security	High
TS-47	NFR-10	Security	Cross-farmer data privacy isolation check	Security	High
TS-48	NFR-03	Reliability	Behaviour on sensor failure	Exceptional	High
TS-49	NFR-03	Reliability	Behaviour on backend API failure	Exceptional	High
TS-50	NFR-03	Reliability	Behaviour on database failure	Exceptional	High
TS-51	NFR-03	Reliability	Behaviour on network interruption	Exceptional	Medium
TS-52	NFR-03	Reliability	System recovery after service restoration	Exceptional	Medium
TS-53	FR-05	Data Acquisition	Valid weather station data ingestion	Positive	Medium
TS-54	FR-06	Data Acquisition	Drone/satellite imagery ingestion and mapping to field	Positive	Medium
TS-55	FR-07	Data Processing	Real-time validation rejects malformed incoming data	Negative	High
TS-56	FR-09	Environmental Monitoring	Continuous soil moisture monitoring trend display	Positive	High
TS-57	FR-10	Environmental Monitoring	Nutrient level monitoring and deficiency flagging	Positive	Medium
TS-58	FR-11	Environmental Monitoring	Irrigation status (on/off) correctly monitored	Positive	High
TS-59	FR-12	Environmental Monitoring	Environmental condition (humidity/temp) monitoring	Positive	Medium
TS-60	FR-22	Reporting	Alert/notification triggered on critical risk event	Positive	High

6. Test Design Techniques

Four complementary test design techniques were applied to AgriTwin's data-driven inputs and workflow logic: Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing and State Transition Testing. Together these techniques improve coverage of both valid/invalid input classes and the platform's stateful data-processing behaviour.

6.1 Equivalence Partitioning

Equivalence Partitioning was applied to key AgriTwin sensor inputs by dividing possible input values into valid and invalid classes, so that one representative value from each partition can stand in for the entire class.

Soil Moisture (%) — Assumed valid range 20–80%

Partition	Description	Representative Value	Expected Handling
EP-1	Valid range	45%	Accepted and processed normally
EP-2	Below valid range	10%	Flagged as out-of-range / rejected
EP-3	Above valid range	95%	Flagged as out-of-range / rejected
EP-4	Missing value	null	Rejected with missing-data error

Partition	Description	Representative Value	Expected Handling
EP-5	Non-numeric value	"ERR"	Rejected with format error

Temperature (°C) — Assumed valid range 5–45°C

Partition	Description	Representative Value	Expected Handling
EP-6	Valid range	28°C	Accepted and processed normally
EP-7	Below expected range	-5°C	Flagged as anomalous reading
EP-8	Above expected range	55°C	Flagged as anomalous reading
EP-9	Invalid format	"twenty-eight"	Rejected with format error

Sensor ID

Partition	Description	Representative Value	Expected Handling
EP-10	Valid registered ID	S-101	Reading accepted and mapped to field
EP-11	Invalid / unregistered ID	S-999	Reading rejected, error logged
EP-12	Empty ID	""	Reading rejected, mandatory-field error
EP-13	Duplicate ID registration	S-101 (already exists)	Registration rejected as duplicate

Representative values are selected to cover one input from each identified partition, minimizing redundant test cases while preserving coverage of all valid and invalid input classes.

6.2 Boundary Value Analysis

Boundary Value Analysis was applied to the soil-moisture input using the same assumed acceptable range (20%–80%) used in Section 6.1. Because AgriTwin's actual threshold configuration was not supplied in the problem statement, these limits are treated as an explicit testing assumption rather than a confirmed system specification.

Test Point	Value	Expected Handling
Minimum – 1	19%	Rejected — below valid range
Minimum	20%	Accepted — lower boundary
Minimum + 1	21%	Accepted — within valid range
Nominal	50%	Accepted — typical value
Maximum – 1	79%	Accepted — within valid range
Maximum	80%	Accepted — upper boundary
Maximum + 1	81%	Rejected — above valid range

Figure 3: Boundary Value Analysis for Agricultural Sensor Data (Soil Moisture)

6.3 Decision Table Testing

A decision table was constructed for the irrigation-recommendation logic, using three assumed binary conditions and four possible recommendation actions.

- C1 = Soil moisture low
- C2 = Temperature high
- C3 = Rain probability low
- A1 = Recommend irrigation A2 = Delay irrigation A3 = Monitor field A4 = Generate alert

Rule	C1: Moisture Low	C2: Temp High	C3: Rain Prob. Low	Resulting Action(s)
R1	Y	Y	Y	A1 + A4 (Irrigate, Alert)
R2	Y	Y	N	A1 (Irrigate)
R3	Y	N	Y	A1 (Irrigate)
R4	Y	N	N	A3 (Monitor field)
R5	N	Y	Y	A3 (Monitor field)
R6	N	Y	N	A3 (Monitor field)
R7	N	N	Y	A2 (Delay irrigation)

Rule	C1: Moisture Low	C2: Temp High	C3: Rain Prob. Low	Resulting Action(s)
R8	N	N	N	A2 (Delay irrigation)

Figure 4: Decision Table for Irrigation Recommendation

6.4 State Transition Testing

State Transition Testing was applied to the Digital Twin's data-processing lifecycle, covering both the successful path from data receipt through recommendation generation, and the exception path taken when incoming data fails validation.

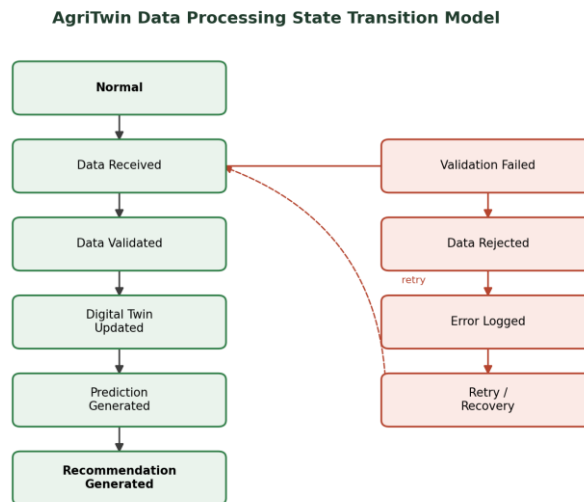


Figure 5: AgriTwin Data Processing State Transition Model

7. Detailed Test Case Design

The test cases below implement the scenarios identified in Section 5, organized by module. As this is a test-case design exercise, no execution has been performed: Actual Result is recorded as "Not Executed" and Pass/Fail Status as "Not Run" for every test case, in accordance with the assignment's academic-integrity requirement. Test data shown represents realistic, illustrative input values for design purposes and does not represent captured sensor readings or system output.

7.1 Authentication Test Cases

TC ID	Req. ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-001	FR-01	Valid login	1. Open login page. 2. Enter valid username/password. 3. Click Login.	user: farmer_raj / pass: Agri@2025	User is authenticated and redirected to dashboard.	Not Executed	Not Run
TC-002	FR-01	Invalid password	1. Open login page. 2. Enter valid username, wrong password. 3. Click Login.	user: farmer_raj / pass: wrongPass1	System displays 'Invalid credentials' and denies access.	Not Executed	Not Run
TC-003	FR-01	Invalid username	1. Open login page. 2. Enter unregistered username. 3. Click Login.	user: unknown_user / pass: Agri@2025	System displays 'User not found' and denies access.	Not Executed	Not Run
TC-004	FR-01	Empty credentials	1. Open login page. 2. Leave both fields blank. 3. Click Login.	user: " / pass: "	System displays mandatory-field validation message.	Not Executed	Not Run
TC-005	FR-01	Multiple failed attempts	1. Attempt login with wrong password 5 times consecutively.	pass: wrong1...wrong5	Account is temporarily locked / warning is displayed after threshold.	Not Executed	Not Run

7.2 Farm and Field Management

TC ID	Req. ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-006	FR-02	Add valid farm	1. Navigate to 'Add Farm'. 2. Enter valid farm details. 3. Submit.	Farm name: Green Valley; Area: 5.2 ha; Location: Lat 11.02, Long 76.96	Farm record is created and listed in farm dashboard.	Not Executed	Not Run
TC-007	FR-02	Missing farm details	1. Navigate to 'Add Farm'. 2. Leave mandatory field(s) blank. 3. Submit.	Farm name: " "; Area: 5.2 ha	System displays mandatory-field validation error.	Not Executed	Not Run
TC-008	FR-02	Duplicate farm	1. Add a farm with a name/ID already registered. 2. Submit.	Farm name: Green Valley (existing)	System rejects duplicate and displays appropriate message.	Not Executed	Not Run
TC-009	FR-03	Update farm details	1. Open existing farm. 2. Modify field(s). 3. Save.	Area updated: 5.2 ha -> 6.0 ha	Updated details are saved and reflected in the record.	Not Executed	Not Run
TC-010	FR-03	Delete farm	1. Select existing farm. 2. Click Delete. 3. Confirm.	Farm ID: F-1001	Farm record is removed and no longer listed.	Not Executed	Not Run

7.3 IoT Sensor Data

TC ID	Req. ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-011	FR-04	Valid sensor data	1. Sensor transmits reading. 2. System receives and stores it.	Soil moisture: 42%; Sensor ID: S-101; Timestamp: valid	Reading is accepted and stored against the correct field.	Not Executed	Not Run
TC-012	FR-04	Missing sensor value	1. Sensor transmits a reading with a missing value field.	Soil moisture: null; Sensor ID: S-101	System flags/rejects the incomplete reading.	Not Executed	Not Run
TC-013	FR-04	Negative invalid value	1. Sensor transmits a physically invalid negative reading.	Soil moisture: -15%	System rejects the reading as out of valid range.	Not Executed	Not Run
TC-014	FR-04	Extremely high value	1. Sensor transmits an unrealistically high reading.	Soil moisture: 250%	System rejects/flags the reading as out of range.	Not Executed	Not Run
TC-015	FR-04	Non-numeric value	1. Sensor transmits non-numeric payload.	Soil moisture: 'ERR'	System rejects the reading with a data-format error.	Not Executed	Not Run
TC-016	FR-04	Duplicate reading	1. Same reading transmitted twice with identical timestamp.	Sensor ID: S-101; Timestamp: identical (x2)	System detects and discards the duplicate entry.	Not Executed	Not Run
TC-017	FR-04	Delayed sensor data	1. Sensor transmits reading with a timestamp older than the delay threshold.	Timestamp: -45 min from current time	System flags the reading as delayed/stale.	Not Executed	Not Run
TC-018	FR-04	Sensor offline	1. Sensor stops transmitting for an extended period.	Sensor ID: S-101; No data for 60 min	System raises a sensor-offline alert.	Not Executed	Not Run
TC-019	FR-04	Incorrect sensor ID	1. Reading transmitted with an unregistered sensor ID.	Sensor ID: S-999 (unregistered)	System rejects reading and logs unrecognized-sensor error.	Not Executed	Not Run
TC-020	FR-04	Timestamp error	1. Reading transmitted with a future/invalid timestamp.	Timestamp: 2099-01-01	System rejects the reading due to invalid timestamp.	Not Executed	Not Run

7.4 Digital Twin Synchronization

TC ID	Req. ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-021	FR-08	Successful synchronization	1. Valid sensor/weather/drone data available. 2. Trigger twin sync.	Complete field dataset for Field F-1001	Digital Twin reflects updated field state accurately.	Not Executed	Not Run
TC-022	FR-08	Partial data synchronization	1. Sync triggered with only some data sources available.	Sensor data present; drone data missing	Twin updates available fields and flags missing sources.	Not Executed	Not Run
TC-023	FR-08	Delayed synchronization	1. Data arrives after the expected sync window.	Data delay: 30 min	Twin updates on arrival and logs the delay.	Not Executed	Not Run
TC-024	FR-08	Conflicting data	1. Two sources report conflicting values for the same parameter.	Sensor: 40% moisture; Satellite estimate: 55%	System applies conflict-resolution rule and logs the discrepancy.	Not Executed	Not Run
TC-025	FR-08	Missing historical data	1. Sync triggered for a new field with no historical data.	Field F-1050 (newly registered)	Twin initializes with available real-time data only.	Not Executed	Not Run
TC-026	FR-08	Synchronization failure	1. Simulate backend unavailability during sync.	Backend service: unreachable	System retries and logs synchronization failure.	Not Executed	Not Run

TC ID	Req. ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-027	FR-08	Recovery after failure	1. Backend restored after failed sync attempt.	Backend service: restored	Twin resumes and completes pending synchronization.	Not Executed	Not Run
TC-028	FR-08	Duplicate data handling	1. Same dataset submitted twice for synchronization.	Field F-1001 dataset (x2)	Duplicate is identified and not double-applied to the twin.	Not Executed	Not Run

7.5 Crop Health Prediction

TC ID	Req. ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-029	FR-13	Valid complete dataset	1. Submit complete sensor/weather/twin data to prediction engine.	Soil moisture, temperature, humidity, NDVI: all present	Prediction engine returns a crop health status.	Not Executed	Not Run
TC-030	FR-13	Missing sensor data	1. Submit dataset with one or more required parameters missing.	NDVI value missing	System handles missing input gracefully (error/partial result).	Not Executed	Not Run
TC-031	FR-13	Insufficient historical data	1. Request prediction for a field with minimal historical records.	Field F-1050; history: 2 days	System returns a low-confidence result or informative message.	Not Executed	Not Run
TC-032	FR-13	Invalid input data	1. Submit malformed/out-of-range input to the prediction engine.	Temperature: -999	System rejects invalid input before prediction.	Not Executed	Not Run
TC-033	FR-13	Prediction service unavailable	1. Simulate prediction service downtime during a request.	Prediction API: unreachable	System displays a service-unavailable message without crashing.	Not Executed	Not Run

7.6 Disease and Pest Prediction

TC ID	Req. ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-034	FR-14	Normal crop condition	1. Submit data representing healthy crop indicators.	NDVI: 0.75; no anomaly flags	System reports normal/no-disease-risk status.	Not Executed	Not Run
TC-035	FR-14	Disease-risk condition	1. Submit data representing disease-risk indicators.	NDVI: 0.32; leaf-stress flag: true	System flags elevated disease risk for the field.	Not Executed	Not Run
TC-036	FR-15	Pest-risk condition	1. Submit data representing pest-risk indicators.	Pest sensor count: high; humidity: elevated	System flags elevated pest-outbreak risk.	Not Executed	Not Run
TC-037	FR-14	Missing input data	1. Submit disease-prediction request with incomplete data.	Leaf-stress flag: missing	System handles missing data without generating a false result.	Not Executed	Not Run
TC-038	FR-15	High-risk alert generation	1. Submit data crossing the defined high-risk threshold.	Pest risk score: above threshold (assumption)	System generates a high-risk alert/notification.	Not Executed	Not Run

7.7 Recommendation Engine

TC ID	Req. ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-039	FR-16	Irrigation recommendation - normal	1. Submit moisture/temperature data within normal range.	Moisture: 45%; Temp: 28°C	System recommends 'Monitor field' (no irrigation action).	Not Executed	Not Run
TC-040	FR-16	Low soil moisture	1. Submit data with soil moisture below assumed threshold.	Moisture: 15% (below assumed min of 20%)	System recommends irrigation.	Not Executed	Not Run
TC-041	FR-16	High soil moisture	1. Submit data with soil moisture above assumed threshold.	Moisture: 85% (above assumed max of 80%)	System recommends delaying irrigation.	Not Executed	Not Run
TC-042	FR-17	Nutrient-deficiency recommendation	1. Submit nutrient data indicating deficiency.	Nitrogen level: low	System recommends appropriate fertilization action.	Not Executed	Not Run
TC-043	FR-18	Pest-risk recommendation	1. Submit data indicating elevated pest risk.	Pest risk: high	System recommends pest-management action.	Not Executed	Not Run

7.8 Dashboard

TC ID	Req. ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-044	FR-19	Dashboard loading	1. Log in. 2. Navigate to dashboard.	Farmer account with 2 active farms	Dashboard loads with correct farm/field summary.	Not Executed	Not Run
TC-045	FR-19	Correct sensor values displayed	1. Open dashboard for a field with recent sensor data.	Field F-1001; latest moisture: 42%	Dashboard displays the correct, current sensor value.	Not Executed	Not Run
TC-046	FR-19	Empty data display	1. Open dashboard for a newly registered field with no data yet.	Field F-1050; no readings	Dashboard displays an appropriate 'no data available' state.	Not Executed	Not Run
TC-047	FR-20	Multiple field visualization	1. Open dashboard for a farmer with multiple registered fields.	Farmer with 3 fields	Dashboard correctly displays data for each field separately.	Not Executed	Not Run

7.9 Predictive Reports

TC ID	Req. ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-048	FR-21	Generate report - valid range	1. Select valid date range. 2. Click Generate Report.	Range: 01-Jul-2026 to 31-Jul-2026	Report is generated with the corresponding field/prediction data.	Not Executed	Not Run
TC-049	FR-21	Invalid date range	1. Select an end date earlier than the start date. 2. Generate.	Start: 31-Jul-2026; End: 01-Jul-2026	System displays a date-range validation error.	Not Executed	Not Run
TC-050	FR-21	Report download/export	1. Generate a valid report. 2. Click Download/Export.	Range: 01-Jul-2026 to 31-Jul-2026	Report file is downloaded/exported in the supported format.	Not Executed	Not Run

7.10 Security

TC ID	Req. ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-051	NFR-02	Unauthorized access attempt	1. Attempt to open a protected module URL without logging in.	Direct URL to /dashboard	Access is denied and user is redirected to login.	Not Executed	Not Run
TC-052	NFR-02	Invalid session	1. Use an expired/invalid session token to call an API endpoint.	Session token: expired	Request is rejected with an authentication error.	Not Executed	Not Run
TC-053	NFR-02	Role-based restriction	1. Log in as a standard farmer role. 2. Attempt to access an admin-only function.	Role: Farmer (non-admin)	Access is denied per role-based access control.	Not Executed	Not Run
TC-054	NFR-02	Malicious input handling	1. Submit a script/SQL-injection style string into an input field.	Input: '<script>alert(1)</script>' / "" OR '1'=1"	Input is sanitized/rejected; no code execution or data breach occurs.	Not Executed	Not Run

7.11 Reliability and Exception Handling

TC ID	Req. ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-055	NFR-03	Network interruption	1. Simulate loss of network connectivity during data submission.	Connectivity: interrupted mid-request	System displays a connectivity error and preserves data integrity.	Not Executed	Not Run
TC-056	NFR-03	Sensor failure	1. Simulate a sensor hardware fault mid-transmission.	Sensor S-101: fault signal	System logs the fault and raises an alert without crashing.	Not Executed	Not Run
TC-057	NFR-03	API failure	1. Simulate backend API returning a server error.	API response: HTTP 500	System displays a graceful error message to the user.	Not Executed	Not Run
TC-058	NFR-03	Recovery after restoration	1. Restore a previously failed service (API/DB/network).	Service state: restored	System resumes normal operation and processes queued data.	Not Executed	Not Run

7.12 Environmental Monitoring and Data Acquisition

TC ID	Req. ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-059	FR-05	Valid weather data ingestion	1. Weather station transmits reading. 2. System receives and stores it.	Temp: 29°C; Humidity: 65%; Rainfall: 2mm; Station ID: W-11	Reading is accepted and linked to the correct field.	Not Executed	Not Run
TC-060	FR-06	Drone/satellite imagery ingestion	1. Drone/satellite imagery uploaded for a field. 2. System maps imagery to field boundary.	Field F-1001; NDVI raster image (valid)	Imagery is accepted and associated with the correct field.	Not Executed	Not Run
TC-061	FR-07	Real-time validation rejects malformed data	1. Backend receives a malformed data packet. 2. Validation layer processes it.	Payload: corrupted JSON	System rejects the packet and logs a validation error.	Not Executed	Not Run
TC-062	FR-09	Soil moisture trend monitoring	1. Open field monitoring view. 2. Review moisture trend over recent period.	Field F-1001; last 7 days of moisture readings	Trend is displayed accurately reflecting stored readings.	Not Executed	Not Run
TC-063	FR-10	Nutrient deficiency flagging	1. Submit nutrient sensor data below the healthy threshold.	Nitrogen: low; Phosphorus: normal; Potassium: normal	System flags nitrogen deficiency on the monitoring view.	Not Executed	Not Run
TC-064	FR-11	Irrigation status monitoring	1. Irrigation system reports on/off status. 2. System records status.	Irrigation valve: ON; Field F-1001	Current irrigation status is correctly displayed on dashboard.	Not Executed	Not Run
TC-065	FR-12	Environmental condition monitoring	1. Environmental sensors report temperature/humidity. 2. System updates monitoring view.	Temp: 33°C; Humidity: 40%	Environmental readings are displayed correctly for the field.	Not Executed	Not Run
TC-066	FR-22	Alert on critical risk event	1. A high-risk prediction/condition is generated. 2. System evaluates alert rule.	Pest risk: high (crosses alert threshold)	System generates and displays a critical alert/notification.	Not Executed	Not Run

8. Requirement Traceability Matrix (RTM)

The Requirement Traceability Matrix maps each functional and safety-critical non-functional requirement to the test scenario(s) and test case(s) that exercise it, ensuring that no significant requirement is left without corresponding test coverage.

Req. ID	Requirement	Scenario IDs	Test Case IDs	Coverage
FR-01	User Authentication	TS-01, TS-02, TS-03, TS-04, TS-05	TC-001, TC-002, TC-003, TC-004, TC-005	Fully Covered
FR-02	Farm Registration	TS-06, TS-07, TS-08	TC-006, TC-007, TC-008	Fully Covered
FR-03	Field Management	TS-09, TS-10	TC-009, TC-010	Fully Covered
FR-04	IoT Sensor Data Collection	TS-11, TS-12, TS-13, TS-14, TS-15, TS-16, TS-17	TC-011, TC-012, TC-013, TC-014, TC-015, TC-016, TC-017, TC-018, TC-019, TC-020	Fully Covered
FR-05	Weather Data Collection	TS-53	TC-059	Fully Covered
FR-06	Drone/Satellite Data Integration	TS-54	TC-060	Fully Covered
FR-07	Real-Time Data Validation	TS-55	TC-061	Fully Covered
FR-08	Digital Twin Synchronization	TS-18, TS-19, TS-20, TS-21, TS-22	TC-021, TC-022, TC-023, TC-024, TC-025, TC-026, TC-027, TC-028	Fully Covered
FR-09	Soil Moisture Monitoring	TS-56	TC-062	Fully Covered
FR-10	Nutrient Monitoring	TS-57	TC-063	Fully Covered
FR-11	Irrigation Status Monitoring	TS-58	TC-064	Fully Covered
FR-12	Environmental Condition Monitoring	TS-59	TC-065	Fully Covered
FR-13	Crop Health Prediction	TS-23, TS-24, TS-25, TS-27	TC-029, TC-030, TC-031, TC-032, TC-033	Fully Covered
FR-14	Disease Prediction	TS-26	TC-034, TC-035, TC-037	Fully Covered
FR-15	Pest Outbreak Prediction	-	TC-036, TC-038	Partially Covered
FR-16	Precision Irrigation Recommendation	TS-28, TS-29, TS-30	TC-039, TC-040, TC-041	Fully Covered
FR-17	Fertilization Recommendation	TS-31	TC-042	Fully Covered
FR-18	Pest Management Recommendation	TS-32	TC-043	Fully Covered
FR-19	Farm Performance Dashboard	TS-33, TS-34, TS-35, TS-36	TC-044, TC-045, TC-046	Fully Covered
FR-20	Crop Growth Visualization	TS-37	TC-047	Fully Covered
FR-21	Predictive Report Generation	TS-38, TS-39, TS-40, TS-41, TS-42	TC-048, TC-049, TC-050	Fully Covered

Req. ID	Requirement	Scenario IDs	Test Case IDs	Coverage
FR-22	Alerts / Notifications	TS-60	TC-066	Fully Covered
NFR-02	Security	TS-43, TS-44, TS-45, TS-46	TC-051, TC-052, TC-053, TC-054	Fully Covered
NFR-03	Reliability	TS-48, TS-49, TS-50, TS-51, TS-52	TC-055, TC-056, TC-057, TC-058	Fully Covered
NFR-10	Privacy	TS-47	-	Partially Covered

Requirement Coverage = (Number of requirements fully covered / Total requirements listed) $\times 100 = (23 / 25) \times 100 = 92.0\%$.

9. Test Coverage Analysis

Total Functional Requirements	22
Total Non-Functional Requirements	10
Total Test Scenarios	60
Total Test Cases	66
Requirements Fully Covered (of listed set)	23 / 25
Requirements Not / Partially Covered	2 / 25
Functional Requirement Coverage	92.0%

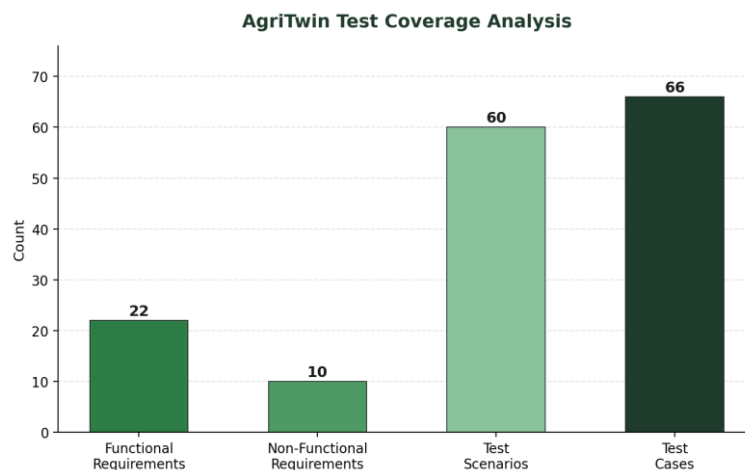


Figure 6: AgriTwin Test Coverage Analysis

Execution-based pass percentage is not calculated because this document represents test-case design rather than executed testing.

10. Security, Reliability and Quality Analysis

10.1 Security

- Authentication of all users prior to accessing farm or field data.
- Authorization and role-based access control between farmer and administrative functions.
- Protection of farmer data privacy, preventing cross-farmer data exposure.
- Input validation to guard against malicious or malformed submissions.
- Secure handling of API communications between the dashboard, backend and prediction services.
- Controlled access to sensor and AI-prediction data to prevent unauthorized retrieval.

10.2 Reliability

- Graceful handling of sensor failures without loss of previously validated data.
- Resilience to network interruptions during data transmission.
- Recovery behaviour following backend API or database failures.
- Correct handling of Digital Twin synchronization failures and retries.

- Fallback behaviour when the AI/ML prediction service is unavailable.
- Logging and monitoring of failure events to support diagnosis and audit.

10.3 Data Quality

- Detection and rejection of missing or incomplete sensor readings.
- Identification and discarding of duplicate data submissions.
- Flagging of invalid or out-of-range sensor readings.
- Handling of timestamp inconsistencies (delayed, duplicate, or future timestamps).
- Resolution of conflicting values reported by different data sources for the same field.

10.4 Usability

- Clarity of dashboard visualizations for non-technical farm users.
- Visibility and comprehensibility of alerts and risk notifications.
- Ease of interpreting crop-health, disease and pest predictions.
- Understandability of irrigation, fertilization and pest-management recommendations.
- Accessibility of the dashboard across supported devices.

10.5 Sustainability

Testing the accuracy and reliability of irrigation and fertilization recommendations directly supports AgriTwin's sustainability goals: correctly triggered irrigation reduces unnecessary water use, validated nutrient recommendations reduce fertilizer waste, and reliable pest/disease alerts help avoid excessive or poorly-timed chemical application. Verifying these behaviours through structured testing is therefore a direct contribution to climate-smart, resource-efficient agriculture.

11. Conclusion and References

11.1 Conclusion

This document has presented a comprehensive test-case design for the AgriTwin Digital Twin platform, covering functional and non-functional requirements, test scenario identification, and detailed test case specification across authentication, farm management, sensor data ingestion, Digital Twin synchronization, AI-based prediction, recommendation generation, dashboard visualization, predictive reporting, security and reliability. The application of Equivalence Partitioning and Boundary Value Analysis strengthens coverage of AgriTwin's numerical sensor inputs by systematically exercising valid and invalid input classes and their boundaries. Decision Table Testing captures the combinational logic underlying irrigation recommendations, while State Transition Testing verifies both the successful data-processing lifecycle and the exception-handling path triggered by validation failures. Together with the Requirement Traceability Matrix, these techniques demonstrate that the designed test cases provide broad, requirement-driven coverage of AgriTwin's functional correctness, data reliability, prediction-workflow behaviour, security, usability and overall quality — establishing a solid foundation for future test execution.

11.2 Assumptions

- Soil moisture and temperature threshold values used in Boundary Value Analysis and the irrigation decision table are testing assumptions, since exact system limits were not specified in the problem statement.
- AI/ML predictions are tested for input/output and error-handling behaviour, not for model accuracy, since no trained model or accuracy results were supplied.
- Actual execution results, screenshots, deployment outcomes and sensor readings are not claimed anywhere in this document; all Actual Result fields are recorded as "Not Executed" and all Pass/Fail Status fields as "Not Run".
- Test data values used throughout Section 7 are illustrative and designed for test-case specification purposes only.

11.3 References

- [1] IEEE Std 829-2008, IEEE Standard for Software and System Test Documentation, IEEE, 2008.
- [2] ISTQB, Certified Tester Foundation Level Syllabus, International Software Testing Qualifications Board, 2018.
- [3] M. Grieves and J. Vickers, "Digital Twin: Mitigating Unpredictable, Undesirable Emergent Behavior in Complex Systems," in Transdisciplinary Perspectives on Complex Systems, Springer, 2017.
- [4] G. J. Myers, C. Sandler, and T. Badgett, The Art of Software Testing, 3rd ed., Wiley, 2011.
- [5] P. Ammann and J. Offutt, Introduction to Software Testing, 2nd ed., Cambridge University Press, 2016.
- [6] R. S. Pressman and B. R. Maxim, Software Engineering: A Practitioner's Approach, 9th ed., McGraw-Hill Education, 2019.